

# Kyrene Chua

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## WORK EXPERIENCE

### Software Engineer

July 2025 – Present

*Defence Science and Technology Agency (Information Program Centre)*

*Singapore*

- Engineered LLM-driven workflows and database architectures for rapid entity resolution and OSINT profiling, utilising advanced prompt engineering to navigate model guardrails.
- Built automated media forensics pipelines to detect threats in the information space, utilizing LLMs, CLIP embeddings, and SerpAPI for advanced image similarity and feature extraction.
- Developed NLP-based bot detection capabilities, analysing narrative embeddings and semantic similarities to identify anomalous behaviours across large datasets.
- Designed data aggregation pipelines and UI dashboards for real-time behavioural analytics (e.g., burst analysis, network interactions), engineering custom data metrics to overcome external API constraints.

### Computer Vision Intern

Jan 2025 - Present

*Polybee*

*Singapore*

- Optimized Python/C++ backend for ZED Camera application, improving performance and system efficiency
- Developed and deployed deep learning pipelines for high-throughput plant phenotyping using multi-sensor data (stereo cameras, NIR, thermal, climate sensors) to analyze plant morphology, health, and detect stress/diseases
- Translated research into scalable industry solutions, achieving production-level accuracy and throughput standards
- Trained and optimized deep learning models for detection and segmentation, enabling real-time deployment on drone-based embedded systems
- Collaborated with cross-functional hardware and software teams to integrate AI models and enhance drone functionality for edge deployment

### AI Developer Intern

May 2024 – November 2024

*SAP SE*

*Singapore*

- Enhanced employee document validation efficiency using Python (Flask, FastAPI), RESTful APIs, OAuth2.0, and OpenAI integration, reducing validation time and improving user satisfaction
- Developed bias detection algorithms for large language models by extracting and integrating bias text guidelines, reducing biased content generation and improving AI output quality
- Maintained and enhanced internal cloud system using Python Flask for secure AI platform consumption (OpenAI, AWS Bedrock, GCP VertexAI, open-source LLMs), increasing system reliability and scalability while ensuring data security and compliance

### Software Engineering Intern

May 2023 – Nov 2023

*Defence Science and Technology Agency (Information Program Centre)*

*Singapore*

- Built data analytics system components using Python, including data collection, database storage, and knowledge representation for departmental analysis
- Integrated ML frameworks (Scikit-learn, PyTorch) and AI modules to enhance analytics capabilities, improving system efficiency and output quality
- Developed Question-Answer system using Generative AI, LLMs, and Prompt Engineering, significantly reducing time required to search company data for information

## EDUCATION

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### National University of Singapore

Aug 2021 - May 2025

*Bachelor of Engineering in Computer Engineering, Honours*

*Singapore*

- Under the Engineering Scholarship Program
- Notable Coursework: CS2040C Data Structures and Algorithms in C++, CS2113 Software Engineering and Object-Oriented Programming, CS3244 Machine Learning, CS2102 Database Systems, EE3801 Data Engineering Principles, CS4225 Big Data Systems for Data Science

## PROJECTS

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### Nomura Trailblazer AI Challenge 2024

- Built a Streamlit app with a resume evaluator, career goals advisor, and course recommendations using AWS to help employees with their career progression
- Implemented Retrieval-Augmented Generation (RAG), vector databases, LLMs, and prompt engineering to enhance information retrieval and user interaction and reduce hallucinations
- Worked in a team of 3 to create and develop our submission and attained 3rd place out of 200 groups

## TECHNICAL COMPETENCIES AND CERTIFICATIONS

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**Languages:** C++/C, Python, Java, SQL, HTML, CSS

**Developer Tools:** Git, Docker, AWS, Azure

**Libraries:** NumPy, Pandas, Matplotlib, scikit-learn, Tensorflow, Pytorch

**Certifications:** DeepLearning.AI Generative AI for LLM, MITx 6.81x Machine Learning with Python